

SARDAR PATEL UNIVERSITY**B.Sc (CA & IT)(3rd - Semester) Examination****Code No: PS03CIIT21 : Computer Programming using C++****Date : 23 -11- 2021, Time :12:00 p.m. to 02:00 p.m.***Tuesday***Note :** Right hand figure indicates the marks of each question.**Total Marks :[70]****Q.1 Multiple Choice Questions****[10]**

- (1) _____ is basic run time entity in object-oriented system.
 - (a) Class
 - (b) Object
 - (c) Data
 - (d) Function
- (2) _____ means "To bind data & functions into a single unit".
 - (a) Inheritance
 - (b) Polymorphism
 - (c) Data hiding
 - (d) Encapsulation
- (3) The header file _____ should be included that use input/output statements.
 - (a) iostream
 - (b) stdio
 - (c) conio
 - (d) io
- (4) _____ is a user define data type.
 - (a) Class
 - (b) Variable
 - (c) Operator
 - (d) Function
- (5) Objects created dynamically with their data members initialized during creation are known as _____ Objects.
 - (a) nameless
 - (b) Live
 - (c) Constant
 - (d) Default
- (6) _____ access specifier is default in class definition.
 - (a) private
 - (b) protected
 - (c) public
 - (d) None of Above
- (7) Creating a new class using one or more existing classes is known as _____.
 - (a) Polymorphism
 - (b) Encapsulation
 - (c) Inheritance
 - (d) Data Hiding
- (8) Which of the following is not a valid argument passing mechanism in C++?
 - (a) Pass by value
 - (b) Pass by address
 - (c) Pass by reference
 - (d) Pass by selection
- (9) Which of the following declarations are illegal?
 - (a) Void *ptr
 - (b) Char *str1 = "xyz"
 - (c) Char str2 = "abc"
 - (d) Const *int p1
- (10) Which one of the following operators can be overloaded?
 - (a) ::
 - (b) New
 - (c) *
 - (d) Sizeof

Q.2 Fill In the blanks & True False

[08]

- (1) _____ is the developer of C++.
- (2) _____ operator is the same as "scanf()" function used in C.
- (3) Constructor is special member function because its name is same as the _____ name.
- (4) One base class & multiple derived classes means _____ inheritance.
- (5) Static binding means that the code associated with a given procedure call at run time. **True Of False**
- (6) Class is a user defines data type. **True Of False**
- (7) In case of multipath inheritance, problem of duplicate data-members is resolved using const keyword. **True Of False**
- (8) A pointer is A variable that stores address of other variable. **True Of False**

Q.3 Short Questions [Attempt any Ten]

[20]

- (1) Define class & object as concept of OOP.
- (2) List out types operator available in C++.
- (3) List out any four header file in C++.
- (4) List the characteristics of member function.
- (5) Define constructor and write characteristics of constructor
- (6) Define classes and objects.
- (7) List situations when inline function is not expanded inline.
- (8) What is a virtual base-class? What is the use of it?
- (9) Define friend function & list its use.
- (10) List out five operators to be overloaded in C++.
- (11) List the operators that can't be overloaded?
- (12) What is **this** pointer?

Q.4 Long Questions [Attempt any Four]

[32]

- (1) What is C++? Explain structure of C++.
- (2) Explain features of OOP in detail.
- (3) Explain all operations on Arrays.
- (4) What do you mean by constructors? What is the use of constructors in C++?
- (5) Explain multiple & multilevel inheritance with suitable example.
- (6) Explain the concept of friend function with suitable example
- (7) Explain read(),write(),seek(),close()
- (8) Explain binary operator overloading with example

